Agenda

- Collection FrameWork
- List

Collection Framework

- Collection framework is Library of reusable data structure classes that is used to develop application.
- Main purpose of collection framework is to manage data/objects in RAM efficiently.
- Collection framework was introduced in Java 1.2 and type-safe implementation is provided in 5.0 (using generics).
- Collection is available in java.util package.
- Java collection framework provides

1. Interfaces -- defines standard methods for the collections.
2. Implementations -- classes that implements various data stuctures.
3. Algorithms -- helper methods like searching, sorting, ...

Collection Hierarchy

- Interfaces: Iterable, Collection, List, Queue, Set, Map, Deque, SortedSet, SortedMap, ...
- Implementations: ArrayList, LinkedList, HashSet, HashMap, ...
- Algorithms: sort(), reverse(), max(), min(), ... -> in Collections class static methods

Collection interface

- Root interface in collection framework interface hierarchy.
- Most of collection classes are inherited from this interface (indirectly).
- Provides most basic/general functionality for any collection
- Abstract methods
  - boolean add(E e)
  - int size()
  - boolean isEmpty()
  - void clear()
  - boolean contains(Object o)
  - boolean remove(Object o)
  - boolean addAll(Collection<? extends E> c)
  - boolean containsAll(Collection<?> c)
  - boolean removeAll(Collection<?> c)
  - boolean retainAll(Collection<?> c)
  - Object[] toArray()
  - Iterator iterator() -- inherited from Iterable
- Default methods
  - default Stream stream()
  - default Stream parallelStream()
  - default boolean removeIf(Predicate<? super E> filter)

List Interface

- Ordered/sequential collection.
- Implementations: ArrayList, Vector, Stack, LinkedList, etc.
- List can contain duplicate elements.
- List can contain multiple null elements.
- Elements can be accessed sequentially (bi-directional using Iterator) or randomly (index based).
- List enables searching in the list
- Abstract methods
    - void add(int index, E element)
    - String toString()
    - E get(int index)
    - E set(int index, E element)
    - int indexOf(Object o)
    - int lastIndexOf(Object o)
    - E remove(int index)
    - boolean addAll(int index, Collection<? extends E> c)
    - ListIterator listIterator()
    - ListIterator listIterator(int index)
    - List subList(int fromIndex, int toIndex)
- To store objects of user-defined types in the list, you must override equals() method for the objects.
- It is mandetory while searching operations like contains(), indexOf(), lastIndexOf().

Assignment

- Q. Create a Employee class and create a List in the main. Test all the functunality of list interface with the employee class. (Note -> Override the equal method when required)